EXECUTIVE SUMMARY:

PREDICTIVE MODELING FOR ALPHA STEEL'S DIGITAL SALES PLATFORM

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OBJECTIVE

USE MACHINE LEARNING MODELS TO PREDICT CLIENT CLASSIFICATION (BROWSER, SELECTOR, QUOTE-ONLY, BUYER).

BUSINESS IMPACT

IMPROVE ALPHA STEEL'S DIGITAL SALES PERFORMANCE BY IDENTIFYING KEY ENGAGEMENT DRIVERS.

DATASET

THIS DATASET SPECIFICALLY FOCUSES ON ALPHA STEEL'S WEB-BASED DIGITAL SALES PLATFORM, WHICH WAS LAUNCHED IN 2016 TO AUTOMATE THE ORDER-TO-PURCHASE CYCLE AND REDUCE RELIANCE ON HUMAN-OPERATED CUSTOMER SERVICE.

CLIENTS INTERACT WITH THE SYSTEM IN DIFFERENT WAYS, LEADING TO FOUR KEY CLASSIFICATIONS:

BROWSER: CLIENTS WHO LOG IN BUT DO NOT SELECT ANY PRODUCTS.
SELECTOR: CLIENTS WHO SELECT PRODUCTS TO CHECK AVAILABILITY BUT DO NOT REQUEST A QUOTE.
QUOTE-ONLY: CLIENTS WHO ADD PRODUCTS TO THE CART AND REQUEST PRICING BUT DO NOT FINALIZE THE PURCHASE.
BUYER: CLIENTS WHO COMPLETE THE PURCHASING PROCESS, SET PAYMENT METHODS, AND CHOOSE PICK-UP OR DELIVERY.

KEY VARIABLES

- Dataset Source: Alpha Steel's web-based sales interactions.
- Predictor Variables:
  - Sessions per year, average actions per session, number of purchases
  - Use of digital tools (Excel, Client Catalogue, Proposal Tool)
  - Order characteristics (tons in cart, tons confirmed, delivery vs. pickup)
  - Customer tenure (months as client, registered months)
- Outcome Variable: Customer classification.
  (Buyer, Selector, Quote-only, Browser)

MODEL PERFORMANCE & COMPARISON

MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-Score	Computation Time	Benefit / Cost Ratio (F1 Score / Computation Time)
Logistic Regression	0.764	0.725	0.764	0.735	1.7 seconds	0.432
Random Forest	0.801	0.777	0.801	0.786	3.2 seconds	0.246
Gradient Boosting	0.814	0.777	0.814	0.783	11.5 seconds	0.068
SVC Linear	0.777	0.732	0.777	0.741	8.0 seconds	0.093

MODEL PERFORMANCE COMPARISON

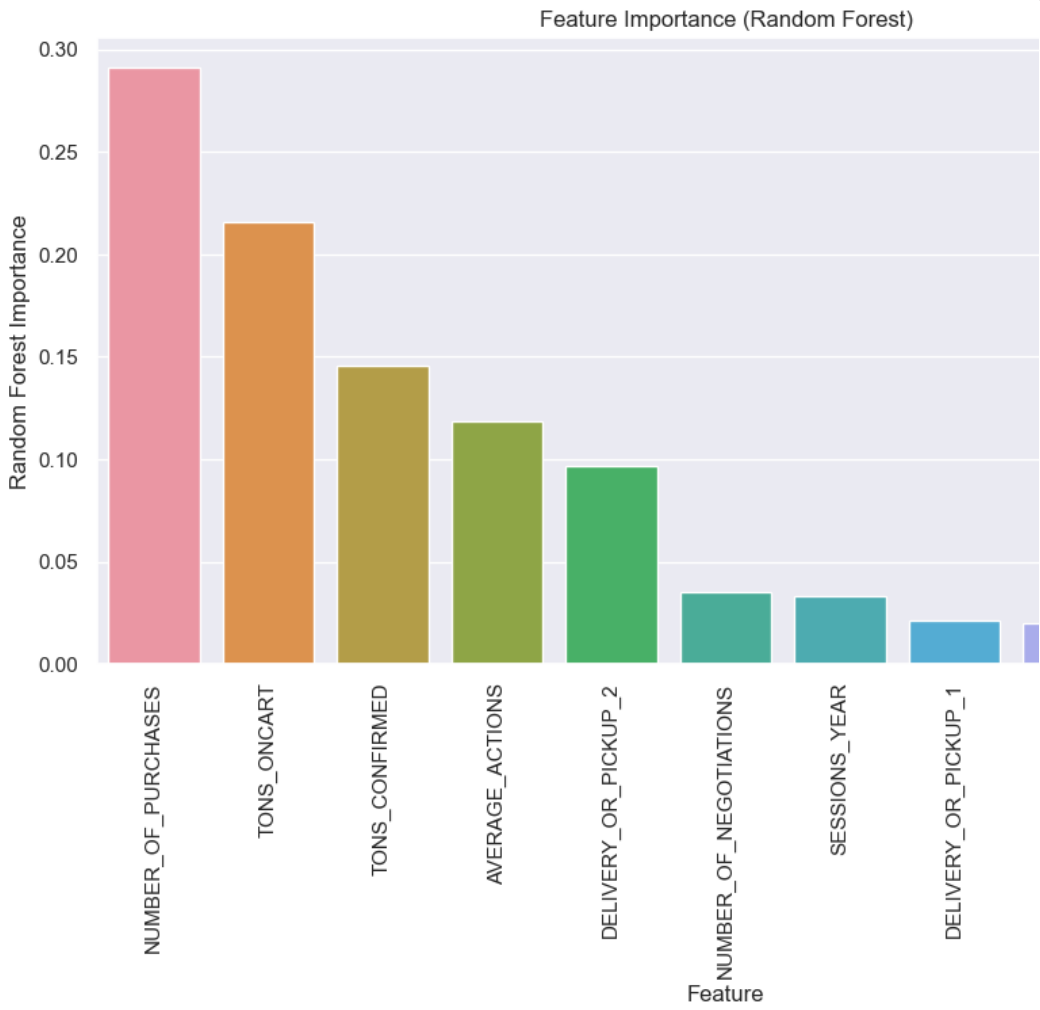
Ranking of Model / BCR

1. Logistic Regression / 0.432
2. Random Forest / 0.246
3. SVC Linear / 0.093
4. Gradient Boosting / 0.068

- Logistic Regression delivers the most benefit per unit of computation time with the highest BCR.
- Random Forest still provides a strong balance between performance and efficiency.
- SVC and Gradient Boosting models are both computationally expensive relative to their performance.

VARIABLE IMPORTANCE RANKING

- <u>Top Predictors (Feature Importance Analysis)</u>
- **Number of Purchases** – Strongest predictor of customer classification
- **Use of Client Catalogue** – Clients using this tool have higher conversion rates.
- **Tonnage in Cart** – Strong correlation with finalizing a purchase.
- **Delivery vs. Pickup** – Preference influences likelihood of completing a purchase.
- **Sessions per year** – Frequent interactions indicate stronger intent.



BUSINESS IMPLICATIONS & RECOMMENDATIONS

ACTIONABLE INSIGHTS:

ENCOURAGE CLIENT CATALOGUE USE – STRONG INDICATOR OF BUYERS.

OPTIMIZE CART EXPERIENCE – CLIENTS WITH MORE TONNAGE IN THEIR CART COVERT BETTER.

PERSONALIZED FOLLOW-UPS – FREQUENT VISITORS WITH UNCOMPLETED PURCHASES SHOULD BE TARGETED.

DELIVERY VS. PICKUP INCENTIVES – IF DELIVERY LEADS TO HIGHER CONVERSION, OPTIMIZE LOGISTICS.

ENHANCE SELF-SERVICE TOOLS – INCREASE AUTOMATION WHERE ENGAGEMENT IS HIGH.

CONCLUSION & NEXT STEPS

KEY TAKEAWAYS:

PREDICTIVE MODELING CAN EFFECTIVELY CLASSIFY AND ENHANCE CUSTOMER ENGAGEMENT

BUSINESS RECOMMENDATIONS ALIGN WITH MODEL INSIGHTS.

FUTURE WORK:

REFINE MODELS AND TEST ADDITIONAL FEATURES FOR GREATER PREDICTIVE ACCURACY

CALL TO ACTION:

IMPLEMENT DATA-DRIVEN STRATEGIES TO ENHANCE ALPHA STEEL'S DIGITAL PLATFORM.

PHOTO CREDITS

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